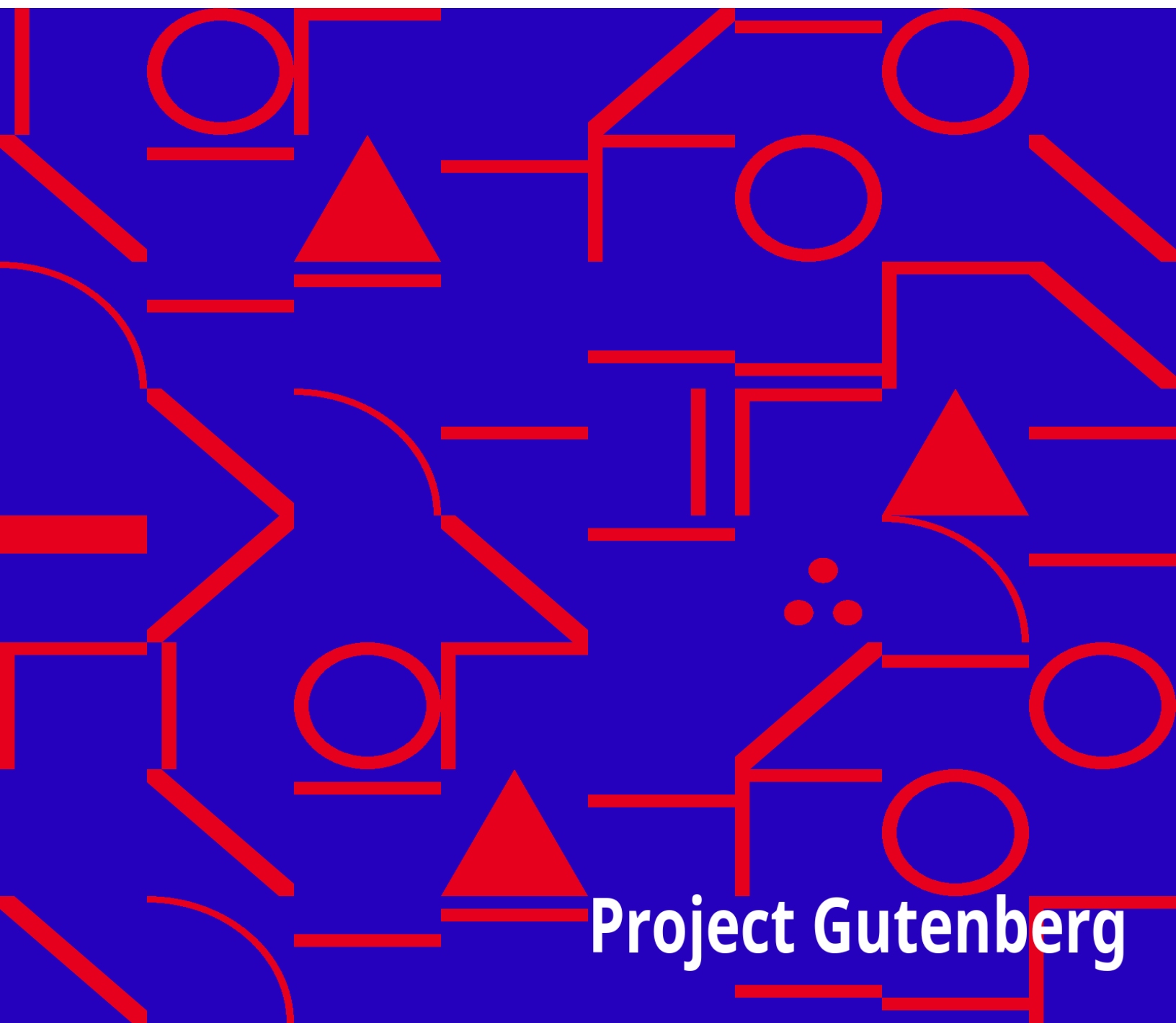


Two New Meadow Mice from Michoacán Mexico

E. Raymond Hall



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Title: Two New Meadow Mice from Michoacán Mexico

Author: E. Raymond Hall

Release date: November 14, 2010 [eBook #34315]

Language: English

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*** START OF THE PROJECT GUTENBERG EBOOK TWO NEW MEADOW MICE FROM MICHOCÁN
MEXICO ***

Two New Meadow Mice from Michoacán Mexico

BY

E. RAYMOND HALL

**University of Kansas Publications
Museum of Natural History**

**Volume 1, No. 21, pp. 423-427, 6 figs. in text
December 24, 1948**

**University of Kansas
LAWRENCE
1948**

UNIVERSITY OF KANSAS PUBLICATIONS, MUSEUM OF NATURAL HISTORY

Editors: E. Raymond Hall, Chairman, A. Byron Leonard, Edward H. Taylor

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**UNIVERSITY OF KANSAS
Lawrence, Kansas**

**PRINTED BY
FERD VOILAND, JR., STATE PRINTER
TOPEKA, KANSAS
1948**

22-3717

Two New Meadow Mice from Michoacán, Mexico

By

E. RAYMOND HALL

In preparing a list of kinds of mammals of which specimens have been saved from the Mexican state of Michoacán, two heretofore unrecognized subspecies of the Mexican meadow mouse, *Microtus mexicanus*, have been found. Names for these and descriptions are given below.

Microtus mexicanus fundatus new subspecies

Type.—Male, adult, skin and skull; No. 100637, Univ. California Mus. Vert. Zool.; 3½ mi. S Pátzcuaro, 7,900 ft., Michoacán, México; March 9, 1943; obtained by E. R. Hall, original no. 5882.

Range.—Known only from the vicinity of the type locality.

Diagnosis.—Size large (see measurements); color brown overlain with cinnamon; nasals expanded distally with premaxillary borders concave laterally; posterior border of orbit inclined posterolaterally; preorbital region and interparietal region depressed; incisive foramina narrow; zygomatic arches parallel; tympanic bullae much inflated.

Comparisons.—Among named subspecies of *Microtus mexicanus*, *M. m. fundatus* most closely resembles *M. m. mexicanus* but differs as follows: Larger in all parts measured; pelage with slightly less buffy color and with the buffy color that is present of a slightly lighter tint; posterior two-thirds of premaxillary border of each nasal concave rather than straight; posterior border of orbit forming more acute angle with sagittal plane of posterior part of skull; superior outline of nasals straight rather than depressed in posterior part; tympanic bullae more inflated both in vertical and horizontal planes.

From *M. c. salvus*, the subspecies next to the westward, *fundatus* differs as follows: Averaging larger in all parts measured; less reddish on upper parts; underparts with more reddish color but the reddish of lighter tint; perineal region buffy instead of plumbeous; nasals with premaxillary borders laterally concave rather than straight; superior outline of skull with nasal segment sloping anteroventrally and interparietal segment sloping posteroventrally rather than straight; posterior margin of orbit inclined posterolaterally thus forming an acute angle, instead of a right angle, with sagittal plane of braincase; zygomatic arches parallel rather than bowed outward; incisive foramina narrower; tympanic bullae more inflated in vertical plane.

Relying on Bailey's (N. Amer. Fauna, 17:55, 1900) description of *Microtus fulviventer*, *fundatus* differs in much larger tympanic bullae.

Remarks.—The series of 59 specimens includes individuals of several ages of both sexes. This has been a great advantage in making comparisons with individuals of geographically adjoining subspecies since individuals of the same age and sex could be compared. When the skulls are laid top-side down on a flat surface the occiput is much higher than in *salvus*.

Our specimens, taken in the dry season, were trapped mostly in runways beneath a dense growth of grass underneath a rail fence.

Specimens examined.—Total, 59, distributed, with respect to the town of Pátzcuaro, as follows: 3½ mi. S, 7,900 ft., 9; 4 mi. S, 7,800 ft., 16; 5 mi. S, 7,800 ft., 26; 9 mi. SE, 8,000 ft., 8.

Microtus mexicanus salvus new subspecies

Type.—Female, adult, skin and skull; No. 52099, Chicago Natural History Museum; Mount Tancitaro, 11,400 ft., Michoacán, México; July 19, 1941; obtained by F. C. Wonder, original No. 1163.

Range.—Known only from Mount Tancitaro at elevations of 7,800 to 11,400 feet.

Diagnosis.—Size medium (see measurements); color brown overlain with cinnamon; premaxillary borders of nasals straight; superior outline of skull nearly straight; posterior margin of orbit at right angle with long axis of skull; zygomatic arches bowed outward; incisive foramina wide; tympanic bullae small.

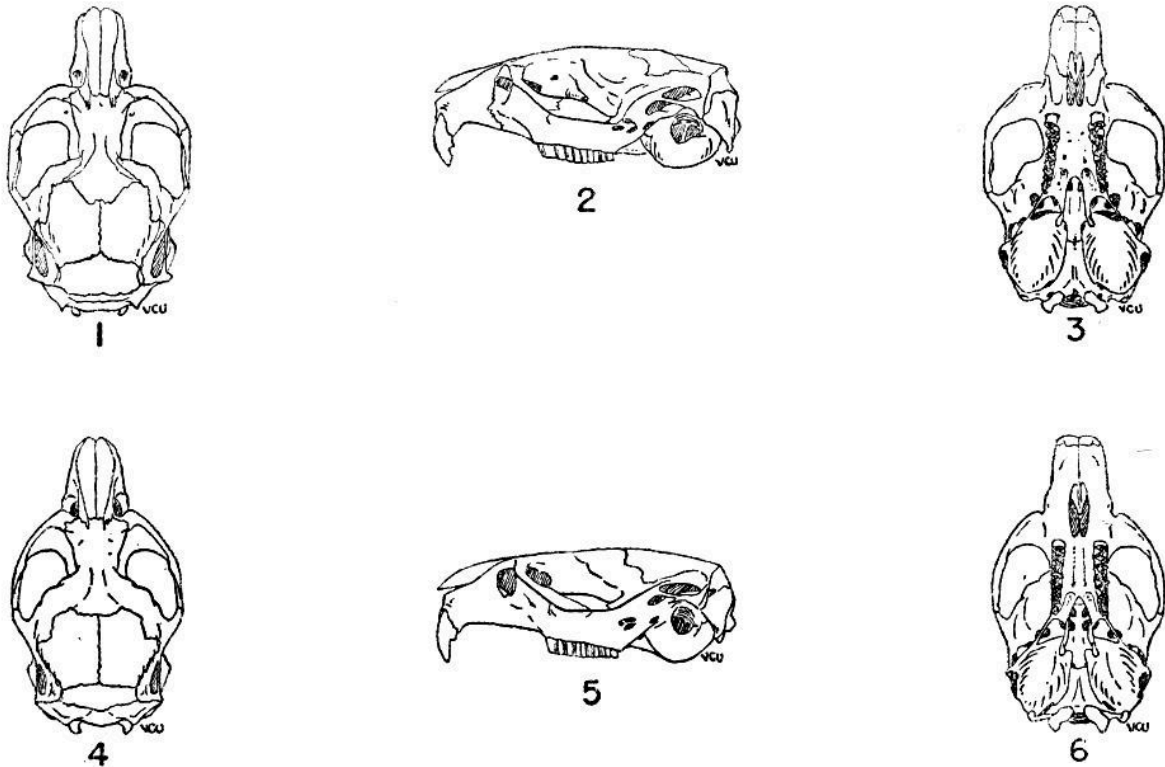
Comparisons.—From topotypes of its nearest relative, *Microtus mexicanus phaeus* Merriam, *salvus* differs as follows: Tail shorter; pelage with slightly more buffy (reddish) pigment which, nevertheless, is of a lighter tint; premaxillary borders of nasals straight rather than concave; superior outline of skull more nearly straight (less convex); inferior border of rostrum more nearly straight (less concave distally); palatal fossae uniformly shallower.

Comparison with *M. m. fundatus* is made in the account of that subspecies.

Remarks.—The degree of difference between *M. m. salvus* and *M. m. fundatus* exceeds that between *M. m. salvus* and *M. m. phaeus* or that between *M. m. fundatus* and *M. m. mexicanus*.

Specimens examined.—Total, 14, from Mount Tancitaro, Michoacán, distributed, according to elevation above sea level, as follows: 11,400 feet, 8; 11,000 ft., 2; 7,800 ft., 1; no elevation recorded, 8.

Measurements of the two subspecies.—Average and extreme measurements of ten adult males of *M. m. fundatus* and eight adult males of *M. m. salvus*, are, respectively, as follows: Total length, 147 (135-158), 141 (134-146); length of tail, 33.4 (30-39), 29.6 (25-32); length of hind foot, 21.1 (20-22), 20.0 (18-21); condylobasal length, 27.7 (27.0-29.0), 25.9 (25.5-26.2); occipitonasal length, 27.5 (26.8-28.5), 25.7 (25.3-26.0); length of nasals, 8.0 (7.7-8.4), 7.5 (6.9-8.1); zygomatic breadth, 16.4 (15.9-17.2), 15.0 (14.6-15.3); interorbital breadth, 3.7 (3.5-3.9), 3.5 (3.3-3.6); mastoid breadth, 12.7 (12.2-13.2), 11.7 (11.7-11.7); alveolar length of upper molar series, 7.0 (6.8-7.3), 6.7 (6.5-6.9); width of rostrum, 5.7 (5.1-5.9), 5.3 (5.3-5.3); palatilar length, 13.3 (12.7-14.3), 12.6 (11.9-12.9). Of *M. m. salvus* only two specimens yield data for the first, second, sixth and eighth cranial measurements named above.



FIGS. 1-6. Skulls of the type specimens of two subspecies of *Microtus mexicanus*. XI.

FIGS. 1-3. *Microtus mexicanus fundatus*.

FIGS. 4-6. *Microtus mexicanus salvus*.

FIGS. 1-6. Skulls of the type specimens of two subspecies of *Microtus mexicanus*. XI.

FIGS. 1-3. *Microtus mexicanus fundatus*.

FIGS. 4-6. *Microtus mexicanus salvus*.

For the loan of comparative material I am grateful to Dr. Hartley H. T. Jackson and Mr. Stanley P. Young of the Biological Surveys Collection in the United States National Museum, Messrs. Karl P. Schmidt and Collin C. Sanborn of the Chicago Natural History Museum, and for assistance with the field work to the John Simon Guggenheim Memorial Foundation and to Miss Annie M. Alexander.

Transmitted June 1, 1948.

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